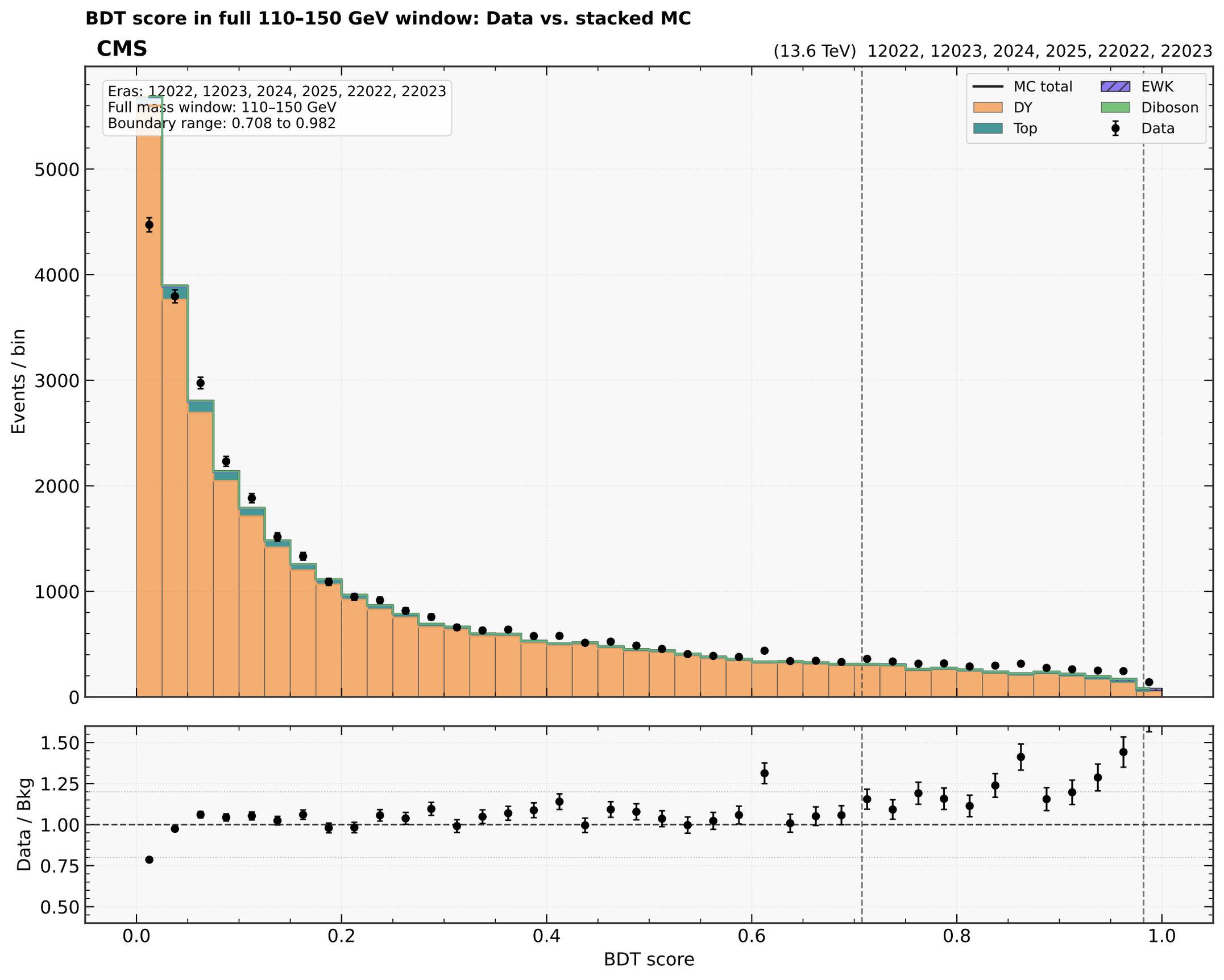
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12022, 12023, 2024, 2025, 22022, 22023
Category edges : 0.0000, 0.7075, 0.9107, 0.9821, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ^2/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_{CB} vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

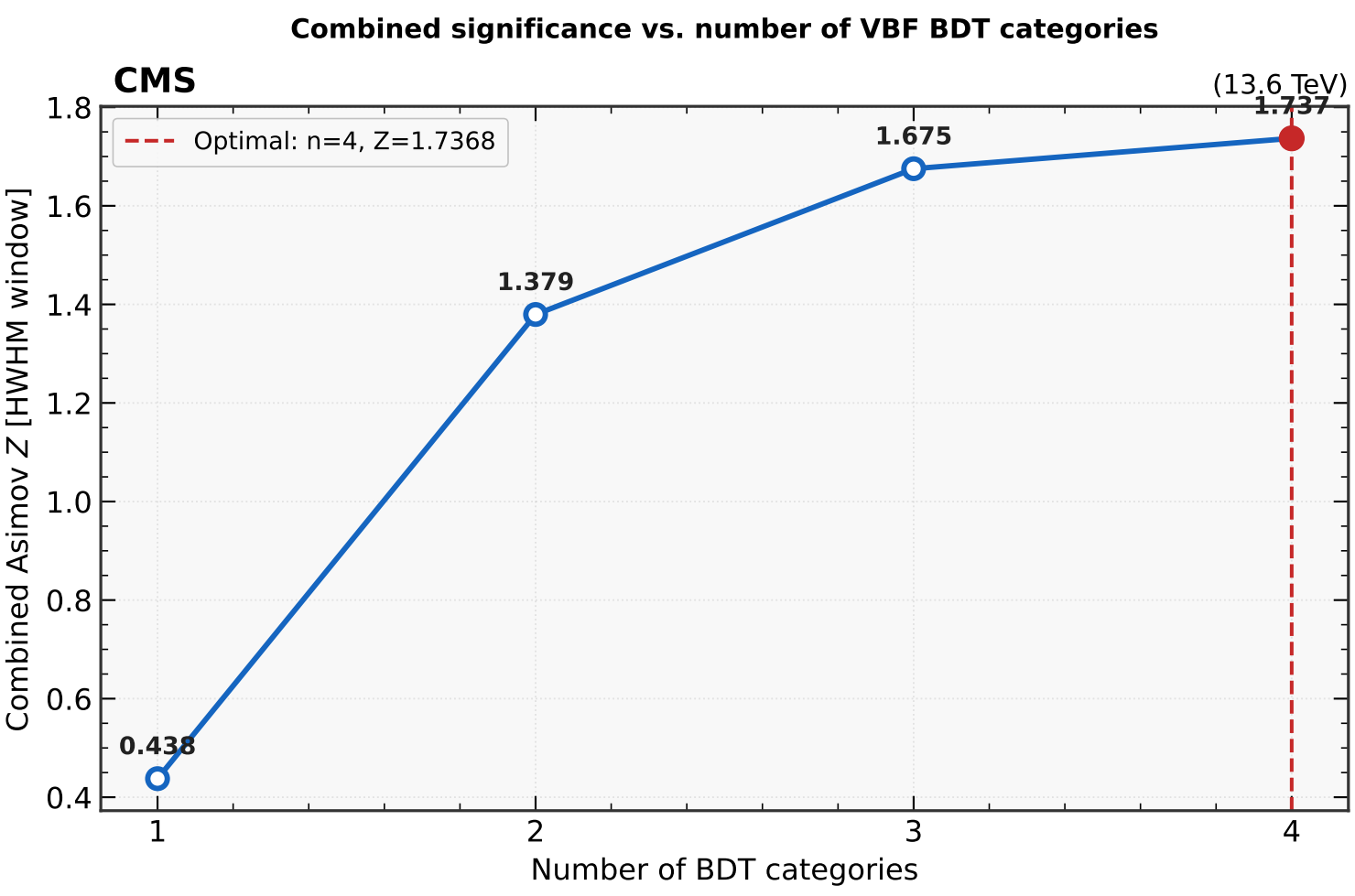

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12022, 12023, 2024, 2025, 22022, 22023
Eras: 12022, 12023, 2024, 2025, 22022, 22023
Full mass window: 110–150 GeV
Boundary range: 0.708 to 0.982
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

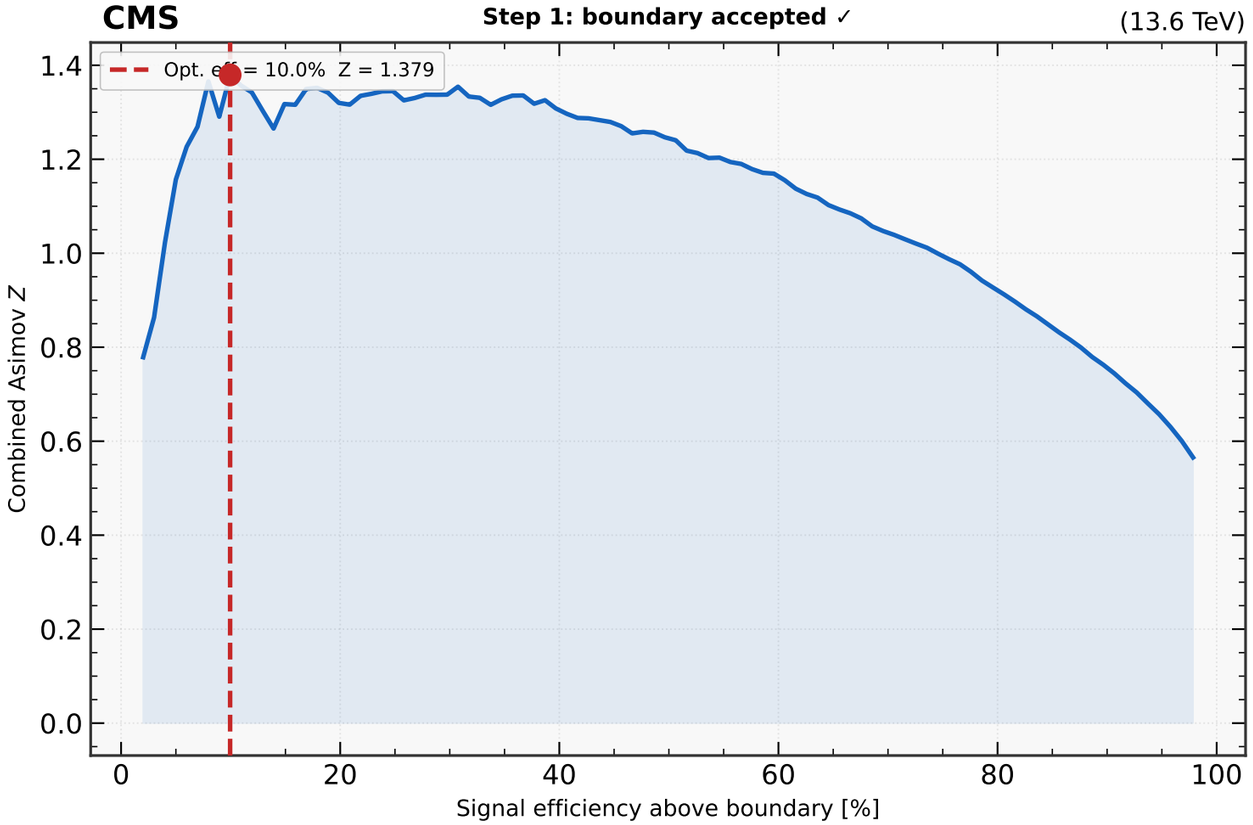
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.185	5688.16 ± 44.78	4472	0.00
[0.025, 0.050)	0.362	3895.40 ± 35.30	3795	0.01
[0.050, 0.075)	0.411	2804.14 ± 29.58	2974	0.01
[0.075, 0.100)	0.451	2137.44 ± 25.65	2231	0.02
[0.100, 0.125)	0.465	1788.63 ± 23.77	1883	0.03
[0.125, 0.150)	0.484	1480.64 ± 21.50	1516	0.03
[0.150, 0.175)	0.505	1255.98 ± 19.63	1332	0.04
[0.175, 0.200)	0.518	1112.88 ± 18.43	1090	0.05
[0.200, 0.225)	0.544	965.14 ± 17.18	948	0.06
[0.225, 0.250)	0.556	867.13 ± 16.35	916	0.06
[0.250, 0.275)	0.575	786.24 ± 15.48	816	0.07
[0.275, 0.300)	0.589	691.76 ± 14.37	758	0.09
[0.300, 0.325)	0.605	665.06 ± 14.09	659	0.09
[0.325, 0.350)	0.636	600.99 ± 13.48	630	0.11
[0.350, 0.375)	0.671	596.65 ± 13.35	638	0.11
[0.375, 0.400)	0.689	530.60 ± 12.55	577	0.13
[0.400, 0.425)	0.709	507.03 ± 12.25	578	0.14
[0.425, 0.450)	0.764	515.18 ± 12.81	513	0.15
[0.450, 0.475)	0.797	479.69 ± 11.99	524	0.17
[0.475, 0.500)	0.830	450.81 ± 11.75	486	0.18

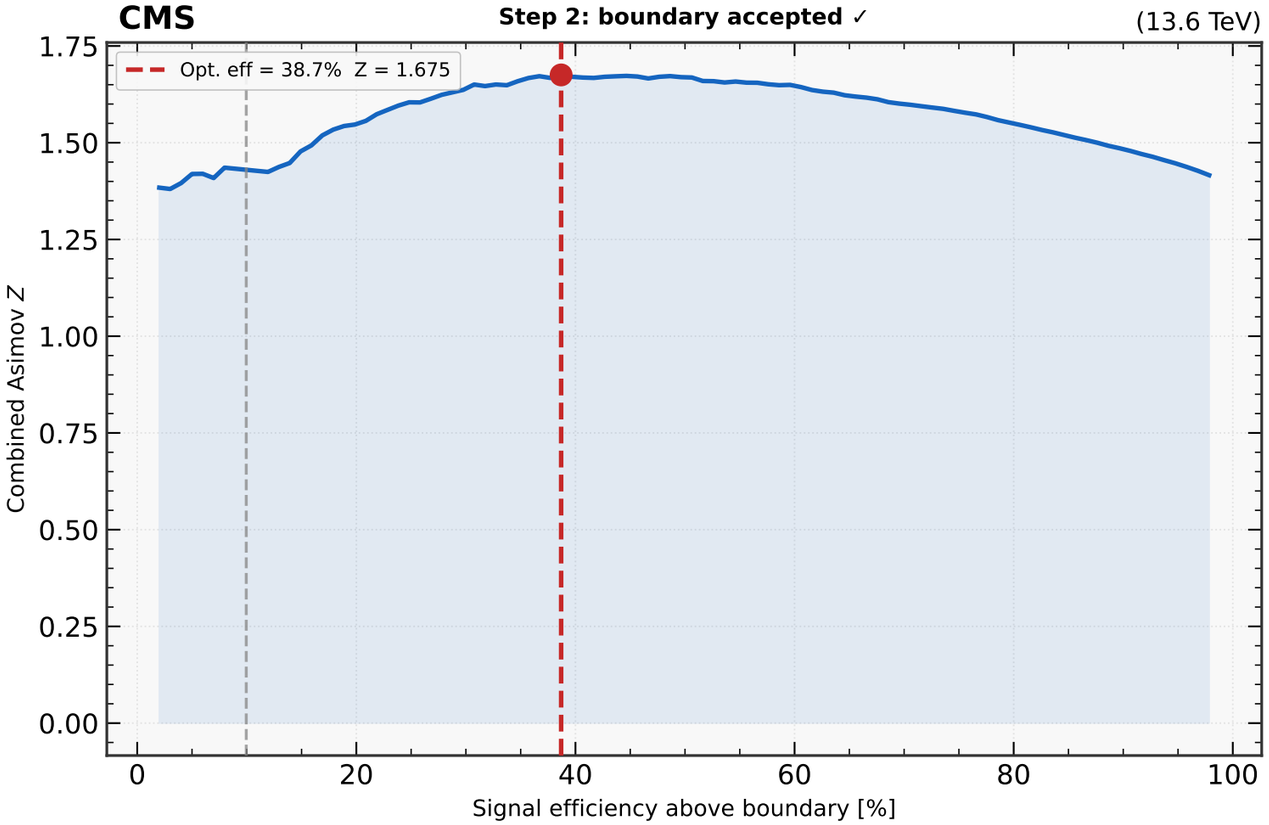
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.884	439.22 ± 11.58	455	0.20
[0.525, 0.550)	0.917	407.21 ± 11.25	406	0.22
[0.550, 0.575)	0.980	380.34 ± 10.63	389	0.26
[0.575, 0.600)	1.036	358.19 ± 10.17	379	0.29
[0.600, 0.625)	1.092	333.72 ± 9.83	438	0.33
[0.625, 0.650)	1.164	337.16 ± 9.84	340	0.34
[0.650, 0.675)	1.235	326.28 ± 9.78	343	0.38
[0.675, 0.700)	1.343	313.11 ± 9.45	331	0.43
[0.700, 0.725)	1.433	312.67 ± 9.48	361	0.46
[0.725, 0.750)	1.562	307.72 ± 9.50	336	0.51
[0.750, 0.775)	1.695	264.46 ± 8.56	315	0.64
[0.775, 0.800)	1.890	274.73 ± 8.95	318	0.68
[0.800, 0.825)	2.074	259.47 ± 8.64	289	0.79
[0.825, 0.850)	2.373	239.86 ± 8.27	297	0.98
[0.850, 0.875)	2.733	223.13 ± 7.79	315	1.21
[0.875, 0.900)	3.249	238.94 ± 8.32	276	1.34
[0.900, 0.925)	3.999	218.89 ± 7.84	262	1.79
[0.925, 0.950)	5.242	194.25 ± 7.30	250	2.63
[0.950, 0.975)	7.586	169.92 ± 6.50	245	4.27
[0.975, 1.000)	9.111	81.87 ± 4.33	140	10.01

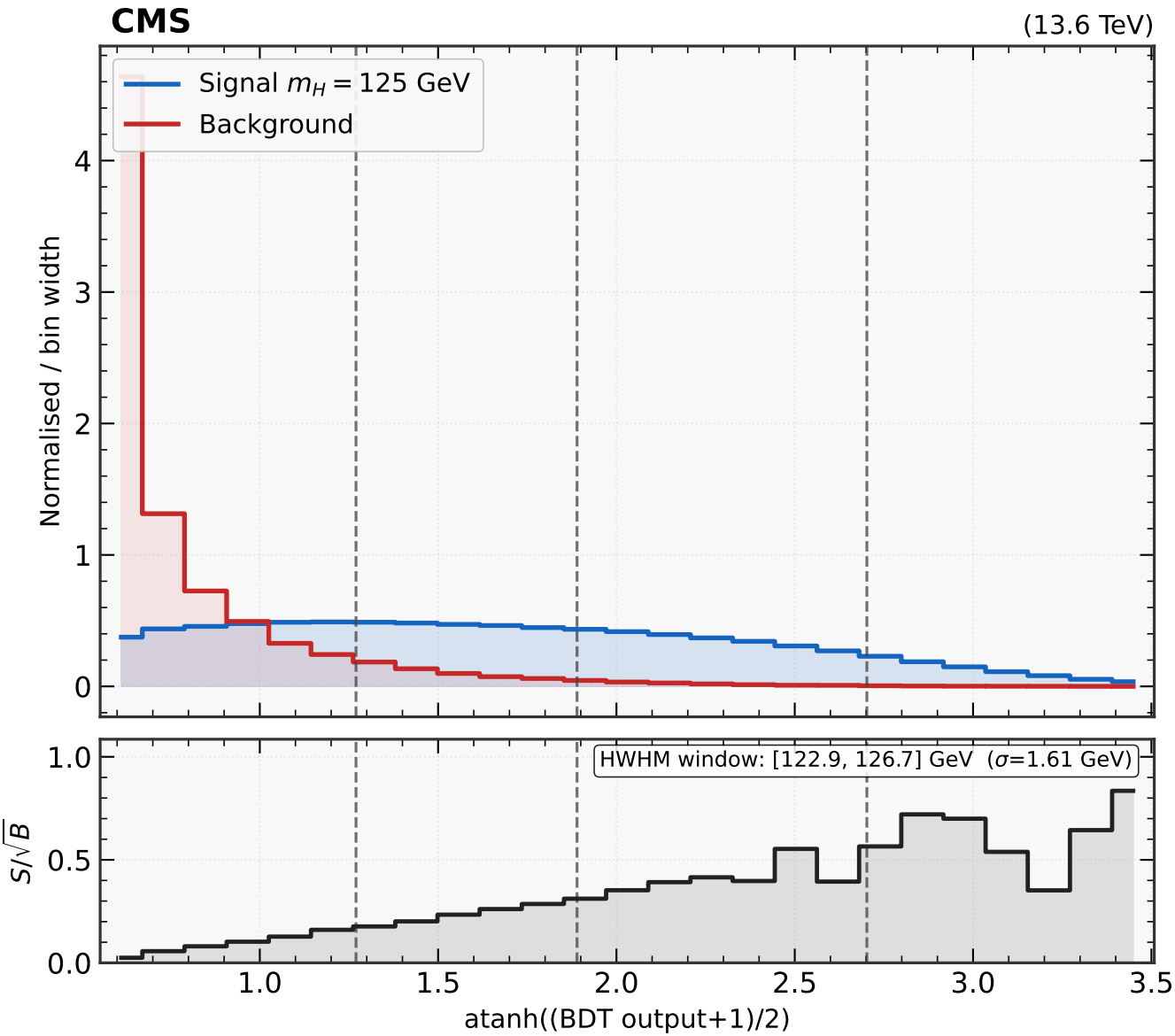

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.7368
0.438
1.379
1.675
1.737
Combined Asimov Z [HWHM window]
Number of BDT categories
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
1
2
3
4

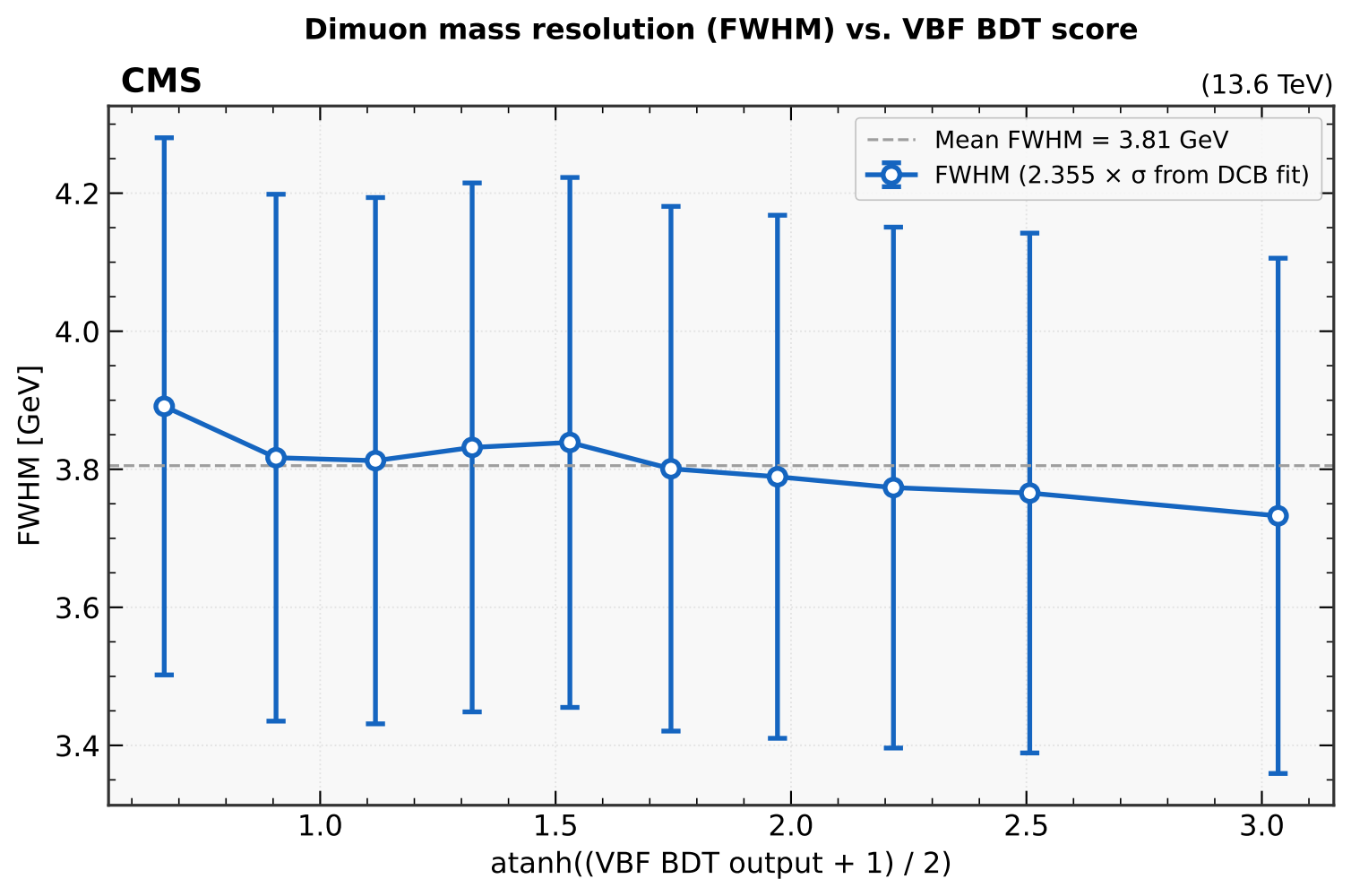
Iterative category boundary optimisation (AN Fig. 32 analog)

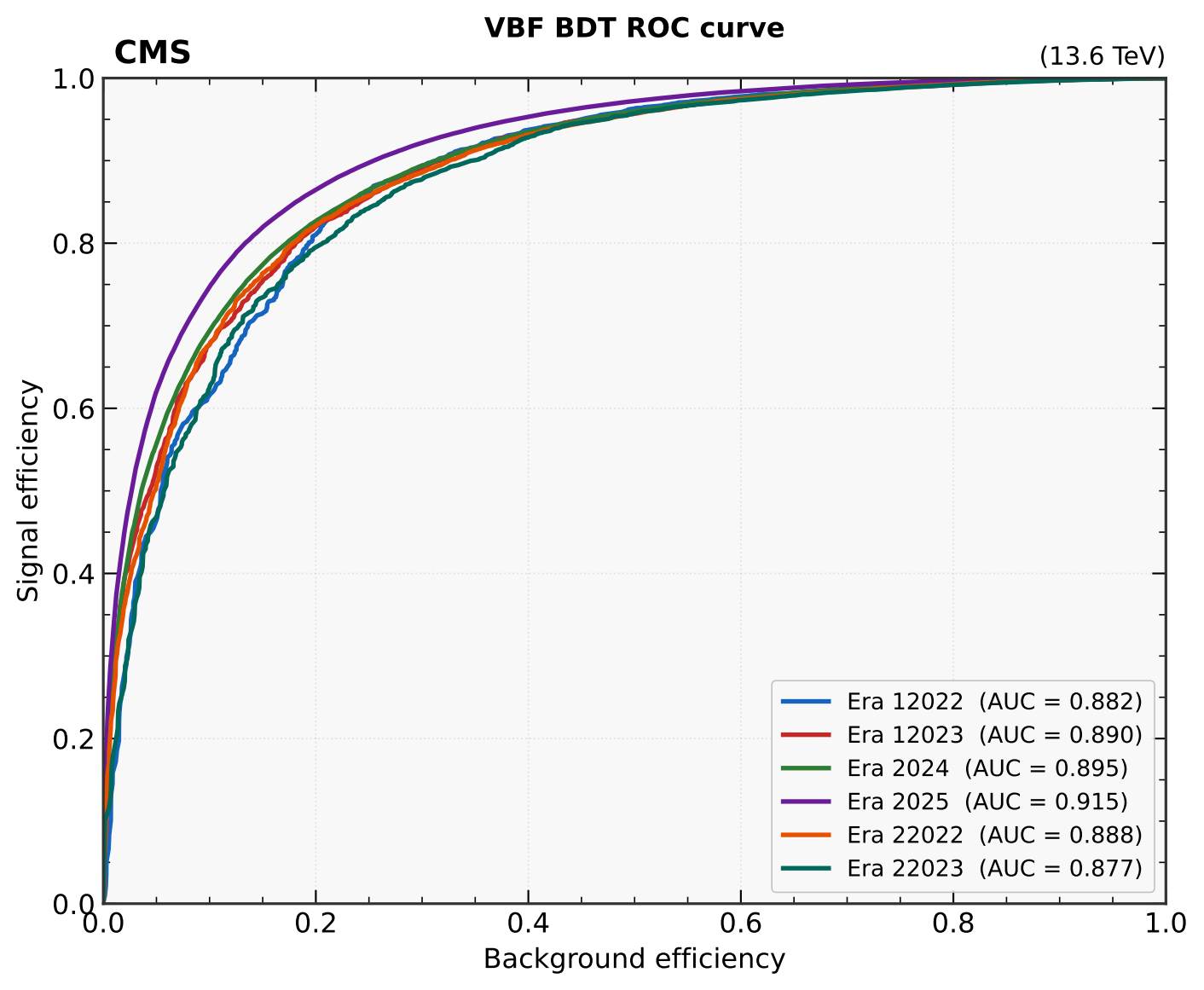


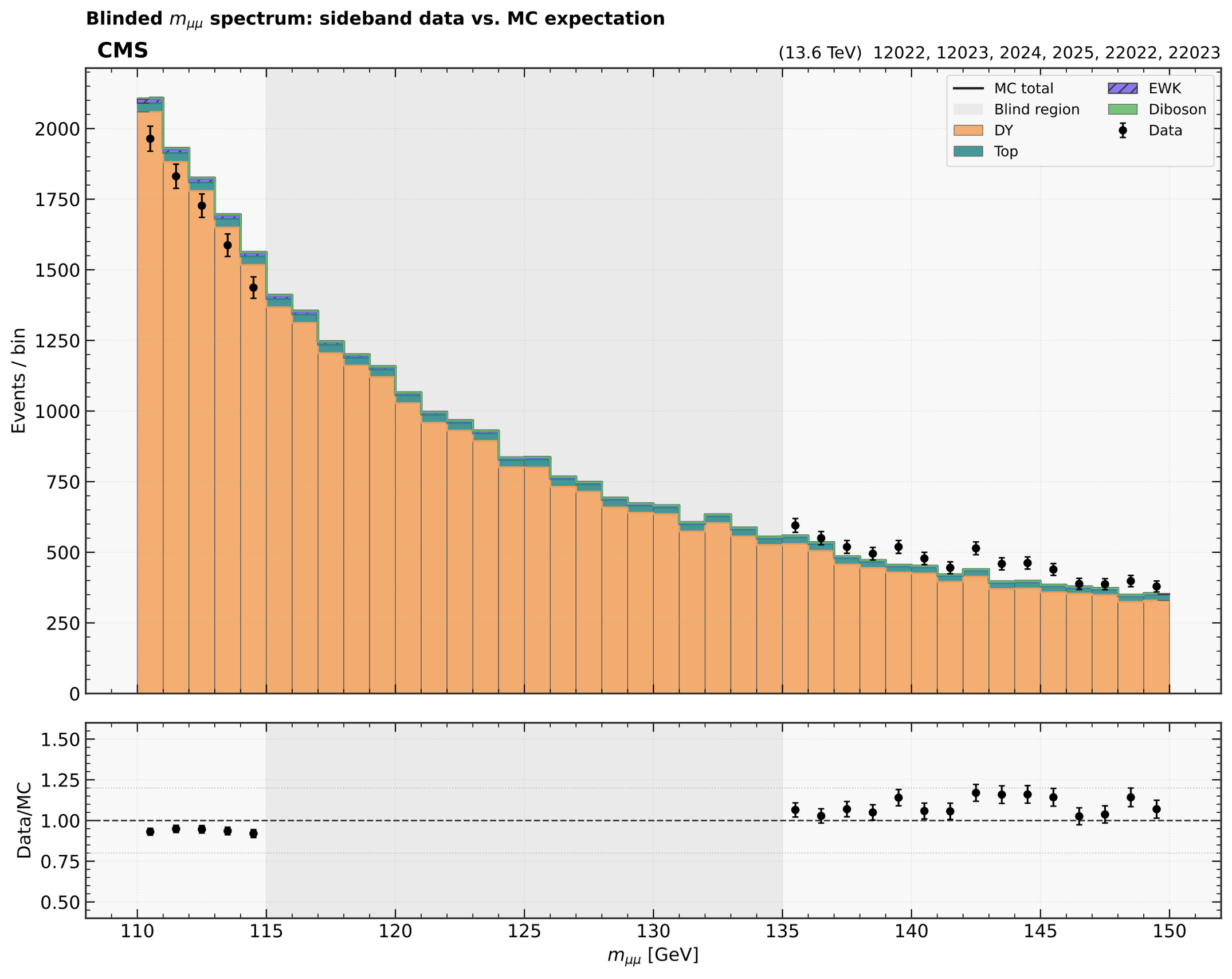


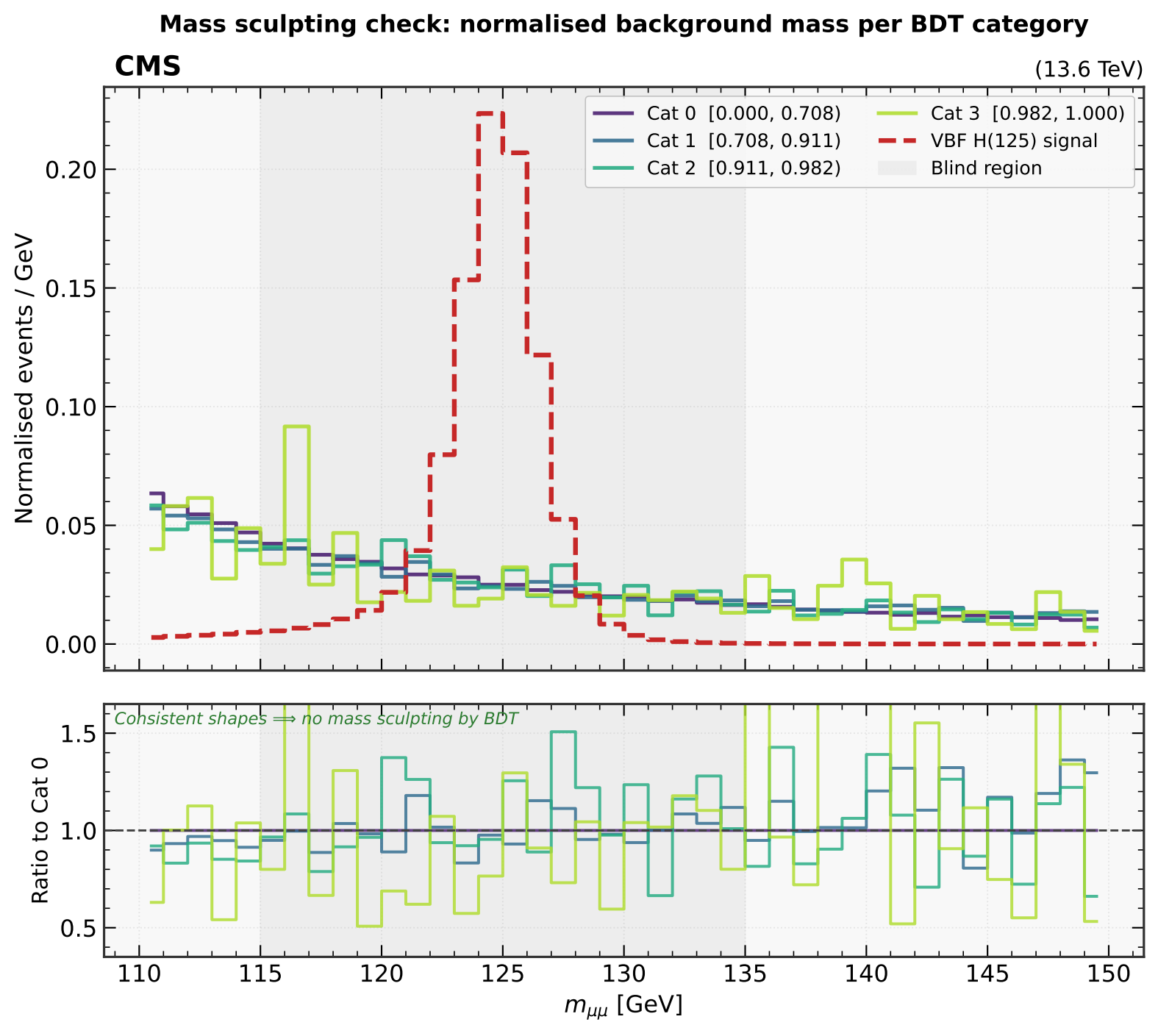
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window



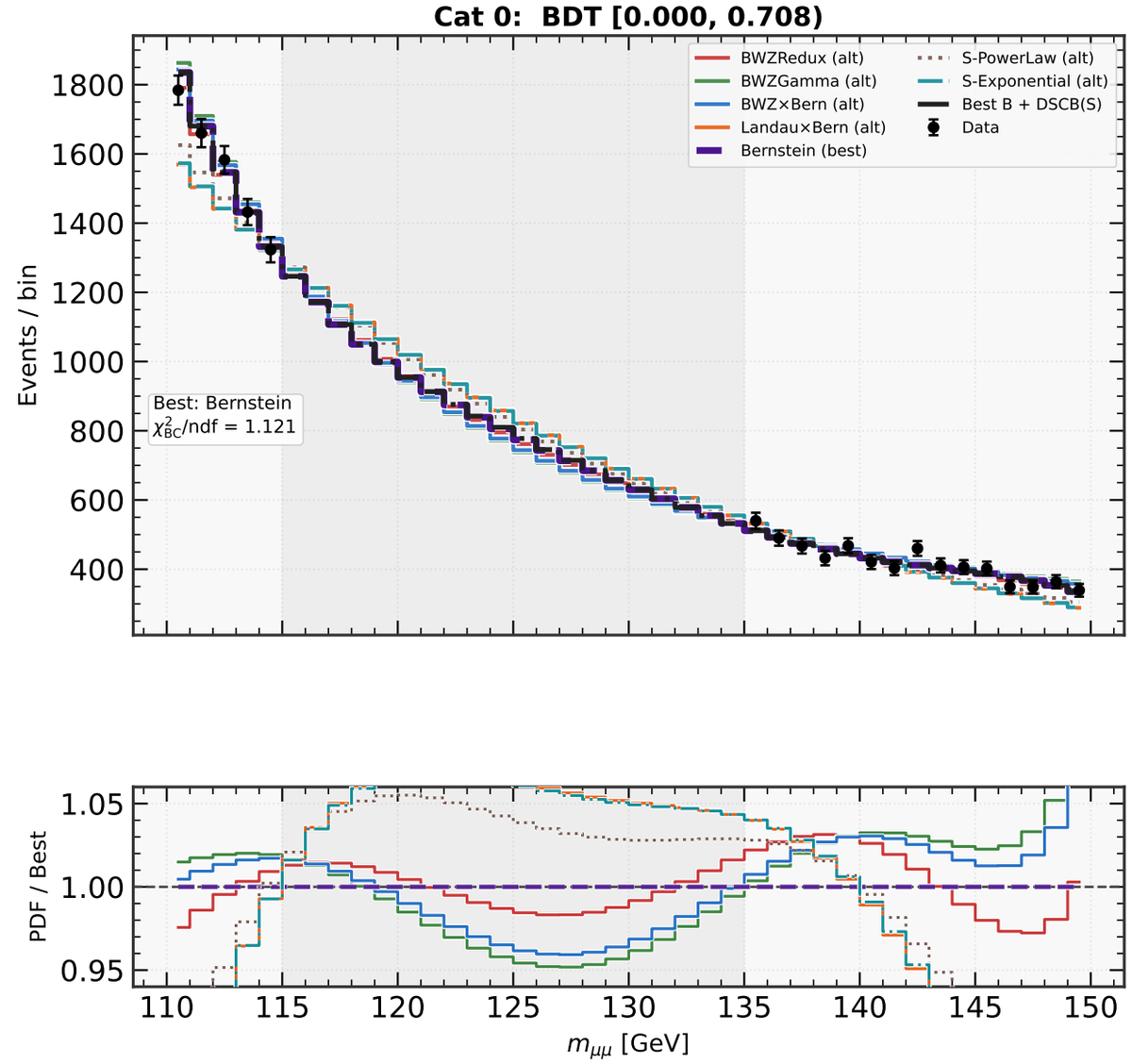

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.81 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

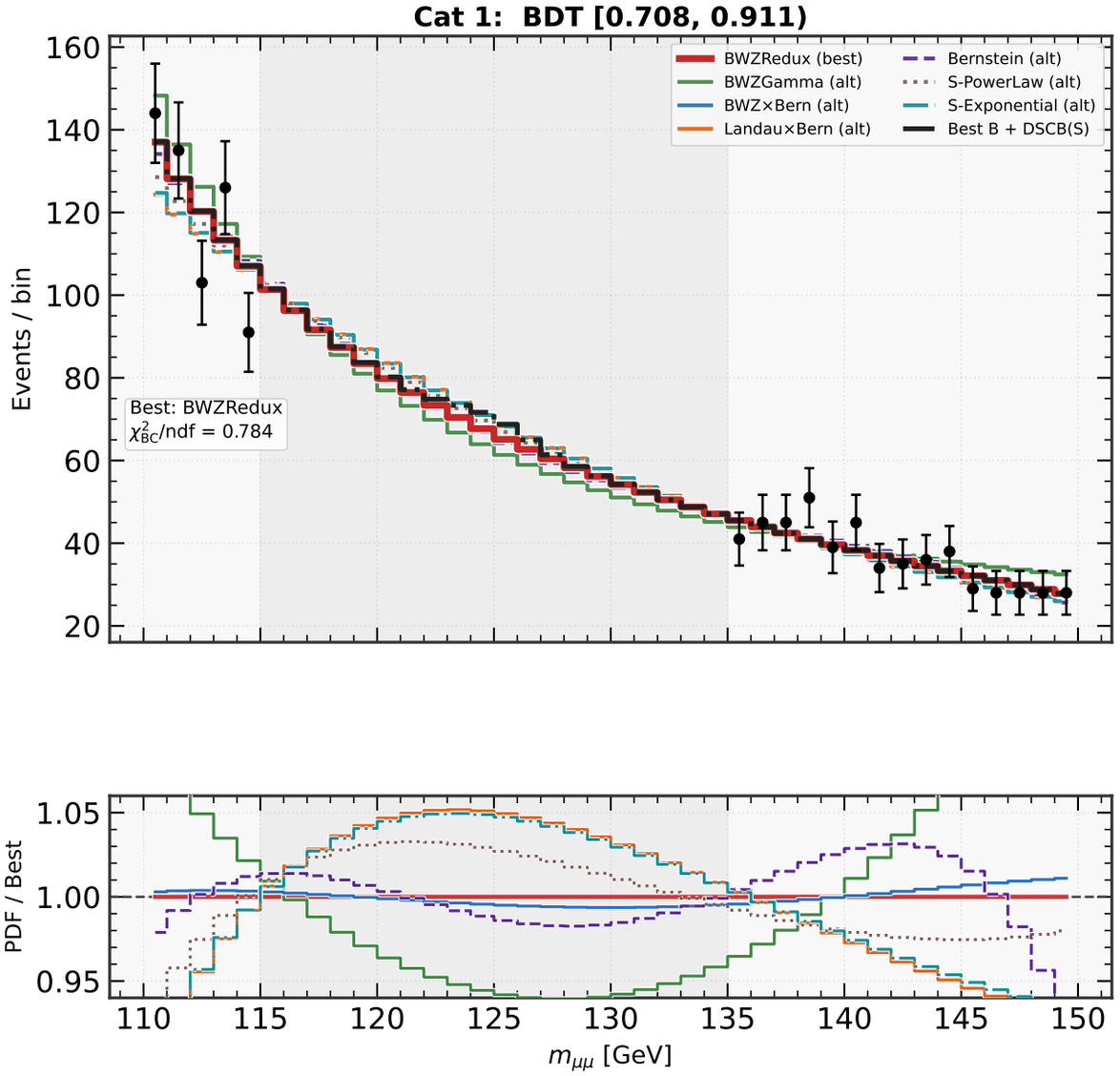

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12022 (AUC = 0.882)
Era 12023 (AUC = 0.890)
Era 2024 (AUC = 0.895)
Era 2025 (AUC = 0.915)
Era 22022 (AUC = 0.888)
Era 22023 (AUC = 0.877)

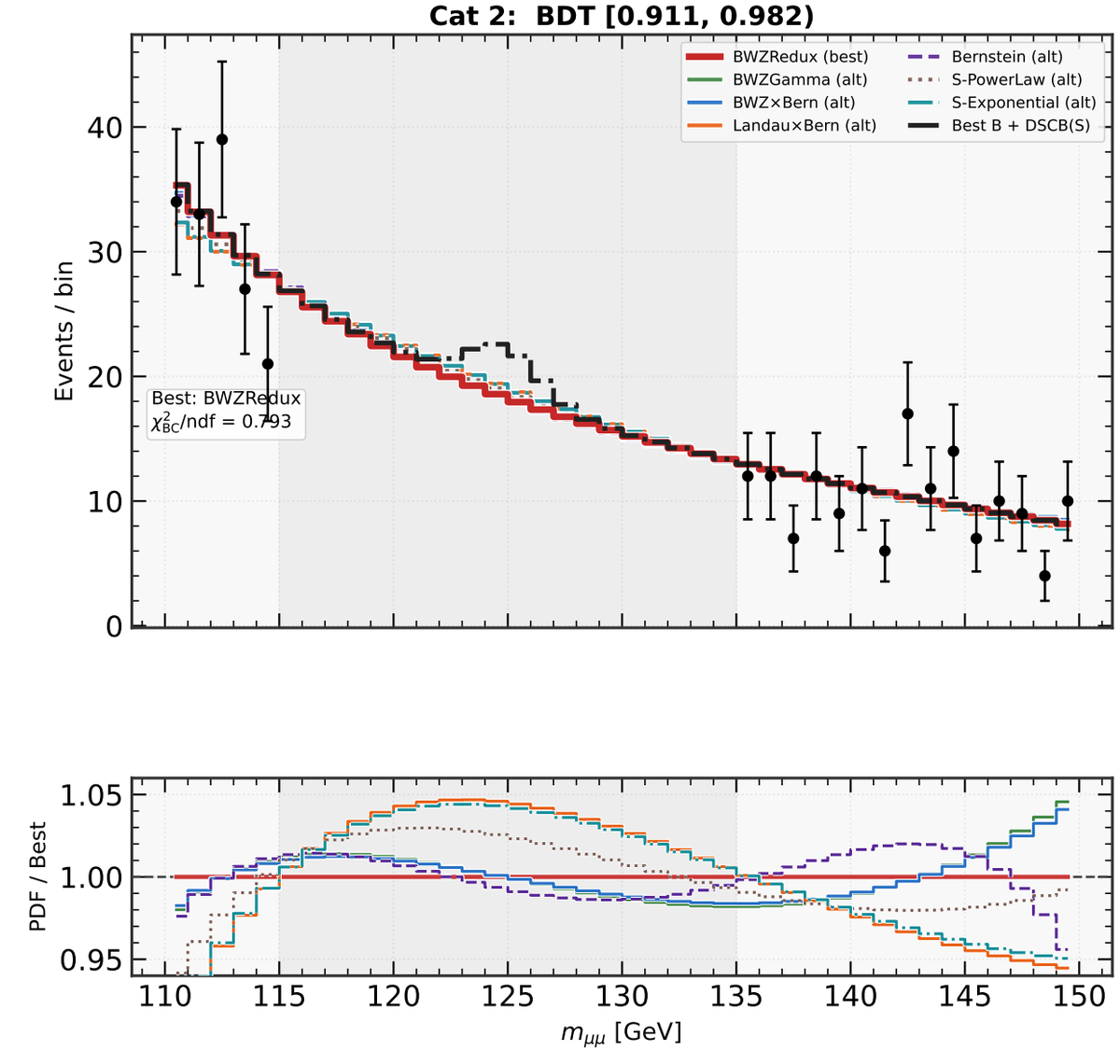

Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12022, 12023, 2024, 2025, 22022, 22023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

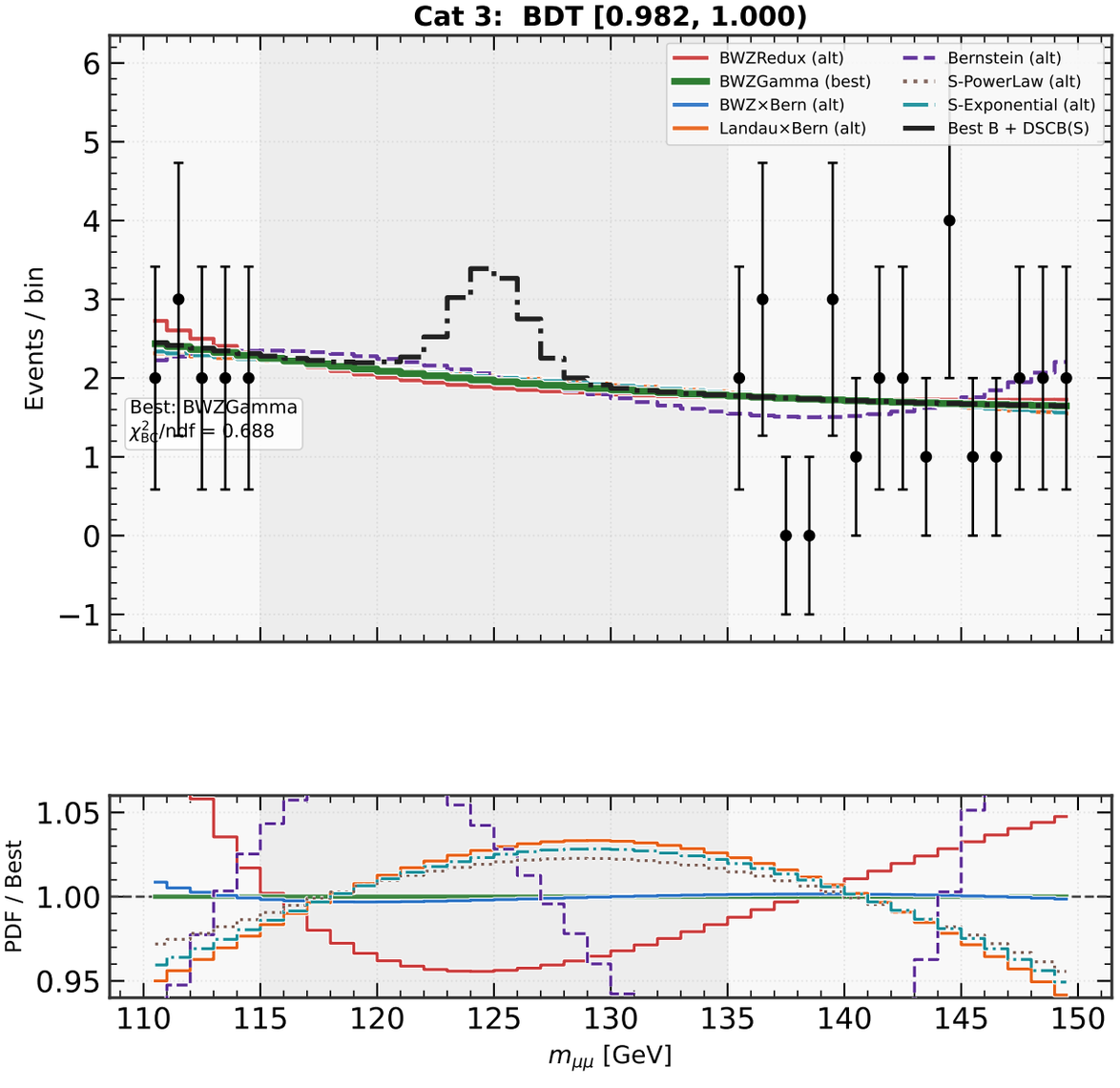

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.708)
Cat 1 [0.708, 0.911)
Cat 2 [0.911, 0.982)
Cat 3 [0.982, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits





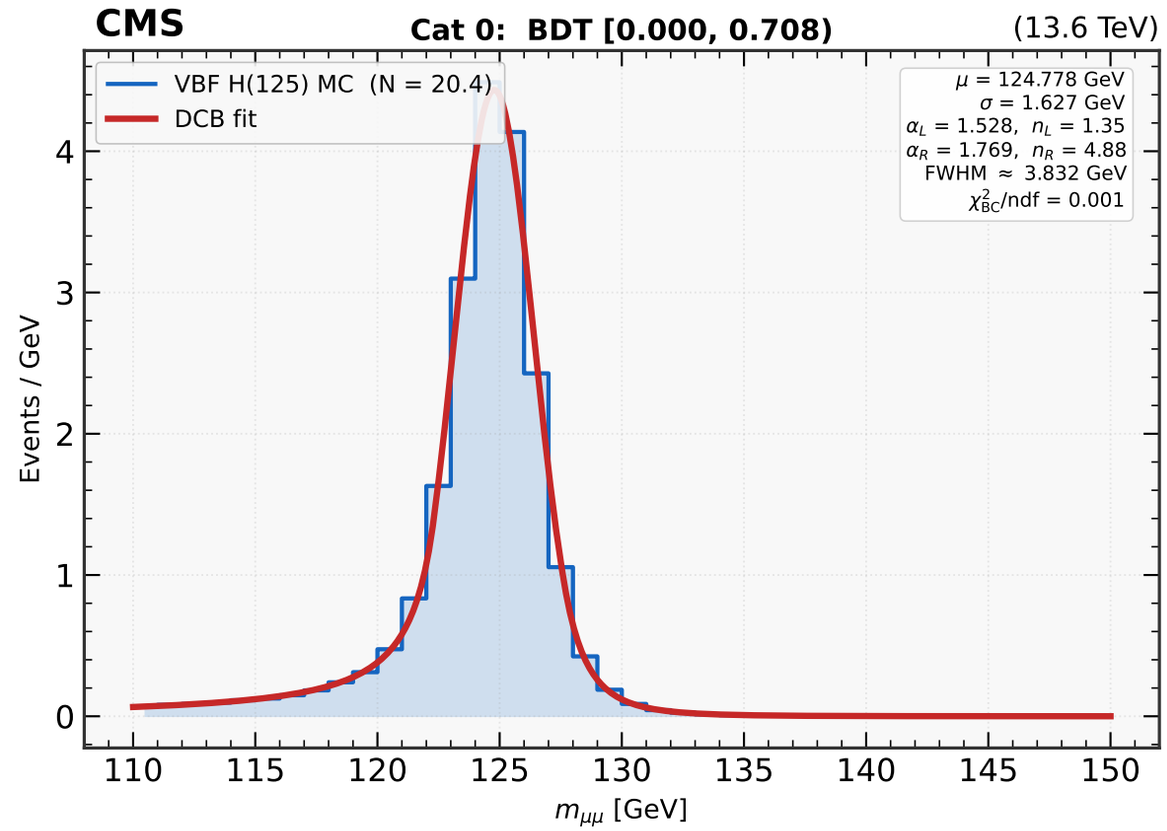


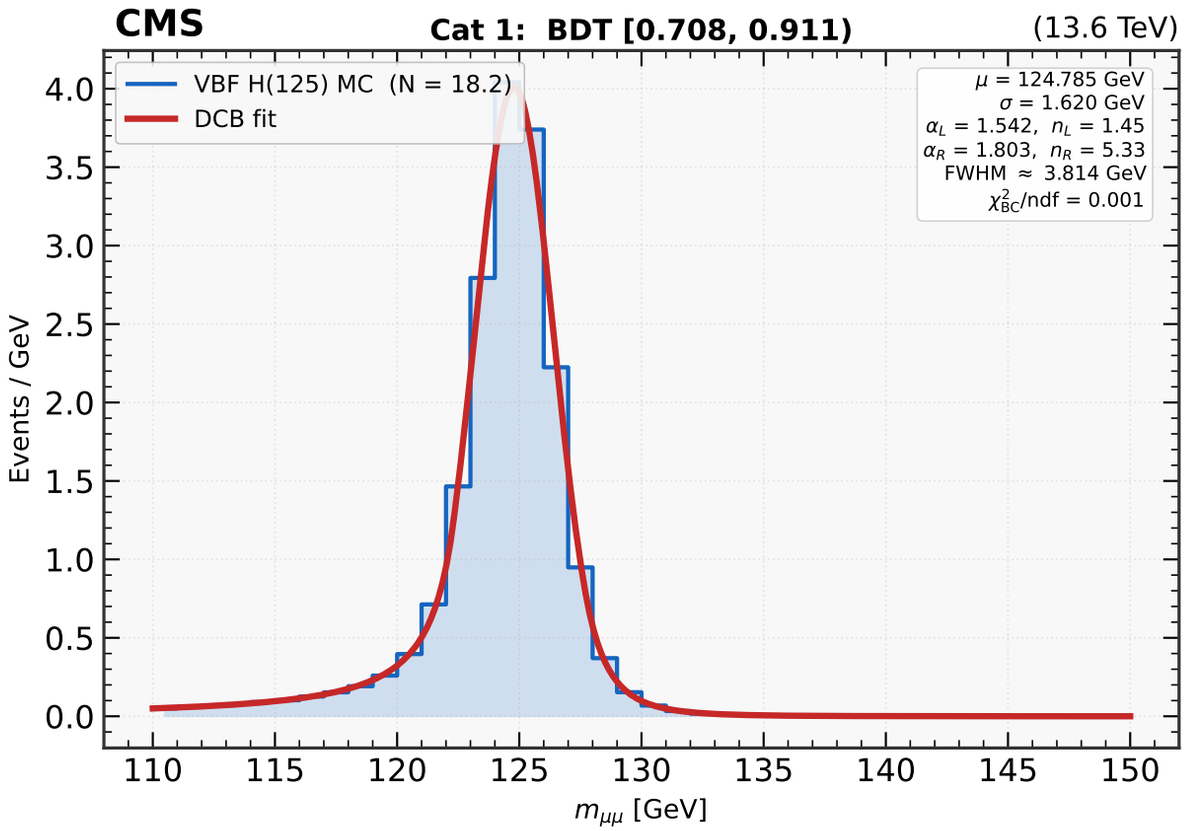


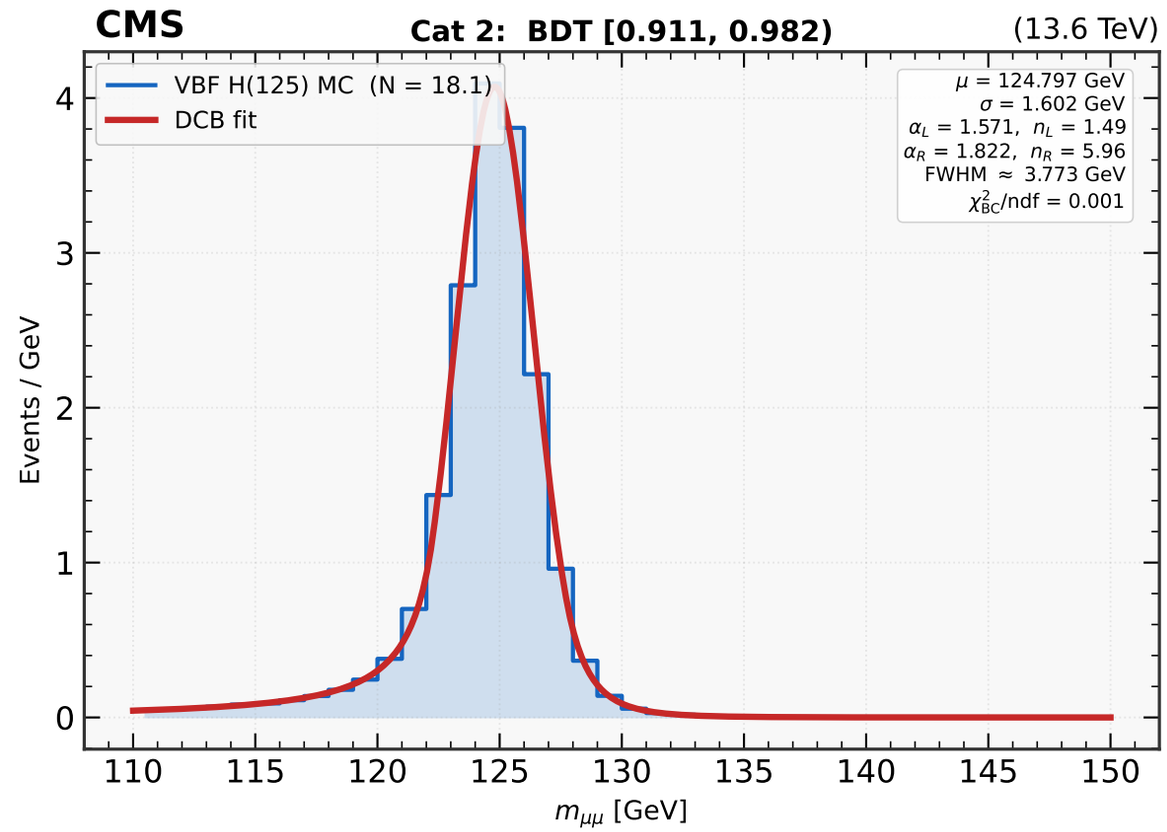
Per-category background model fit comparison (χ²_BC/ndf, ndf)

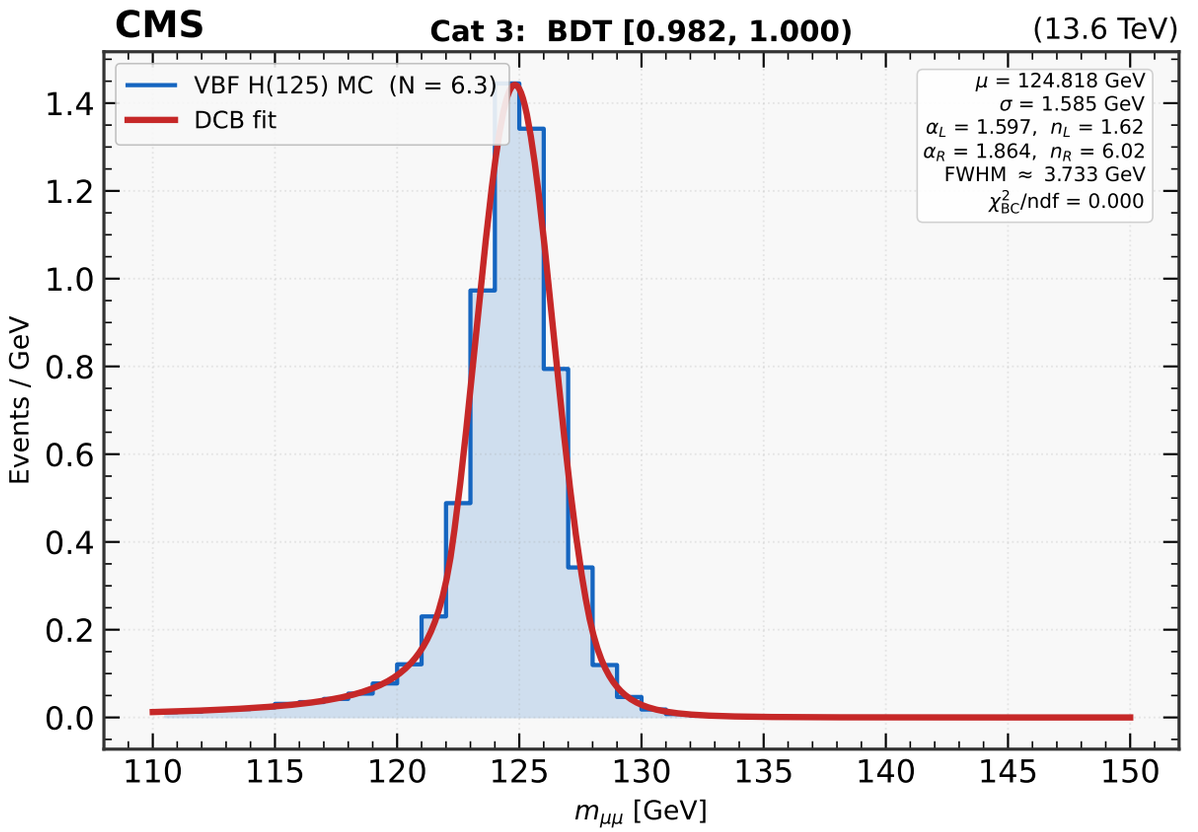
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.708)	Bernstein	1.121	1.211 (37)	1.837 (37)	1.612 (36)	5.041 (37)	1.121 (33)	3.095 (37)	4.859 (37)
1	[0.708, 0.911)	BWZRedux	0.784	0.784 (37)	0.980 (37)	0.831 (35)	0.942 (37)	0.835 (35)	0.853 (37)	0.930 (37)
2	[0.911, 0.982)	BWZRedux	0.793	0.793 (37)	0.796 (37)	0.818 (36)	0.823 (37)	0.839 (35)	0.804 (37)	0.820 (37)
3	[0.982, 1.000)	BWZGamma	0.688	0.693 (37)	0.688 (37)	0.708 (36)	0.694 (37)	0.707 (35)	0.692 (37)	0.693 (37)

VBF H(125) signal DCB extraction per BDT category



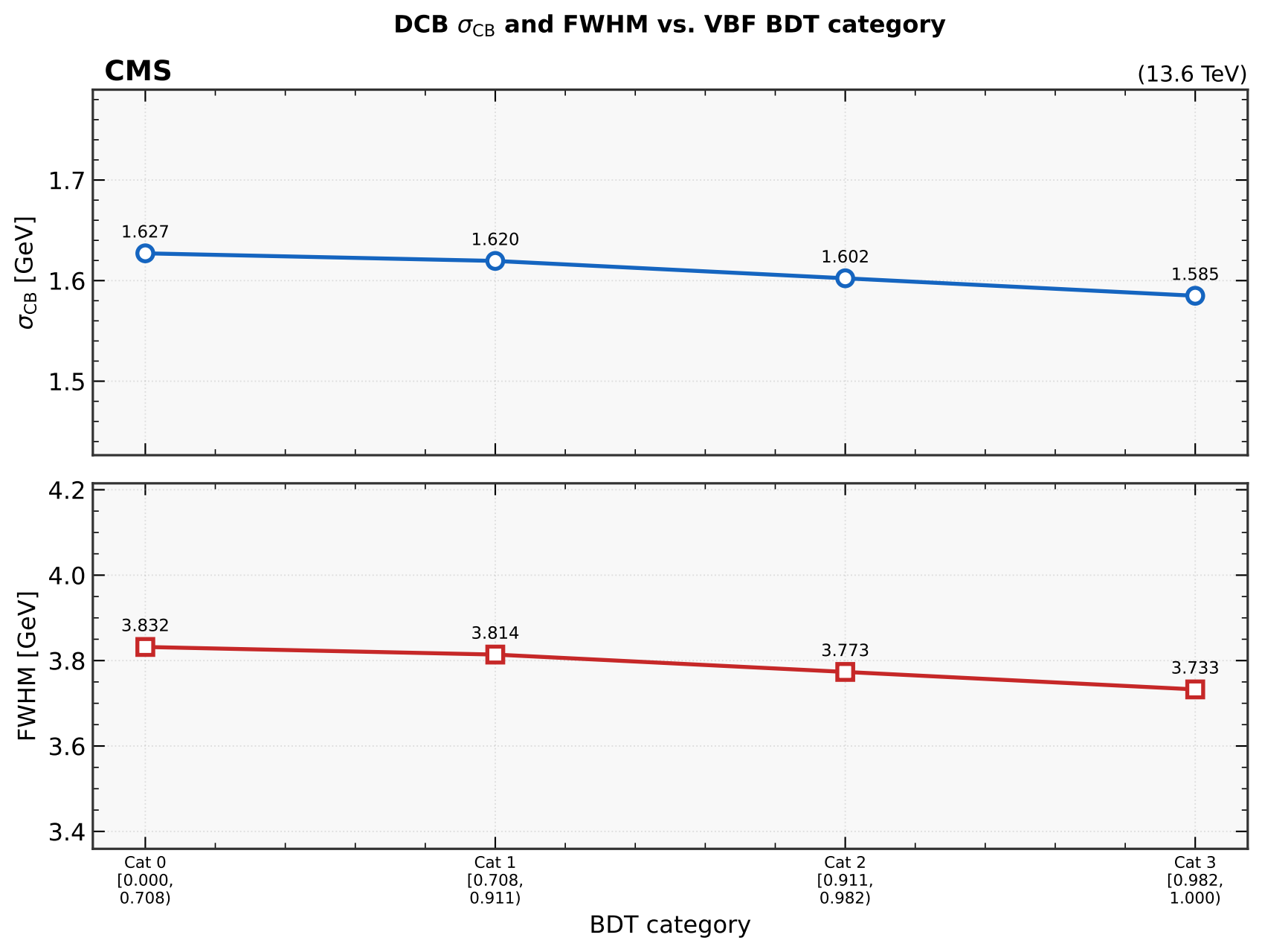


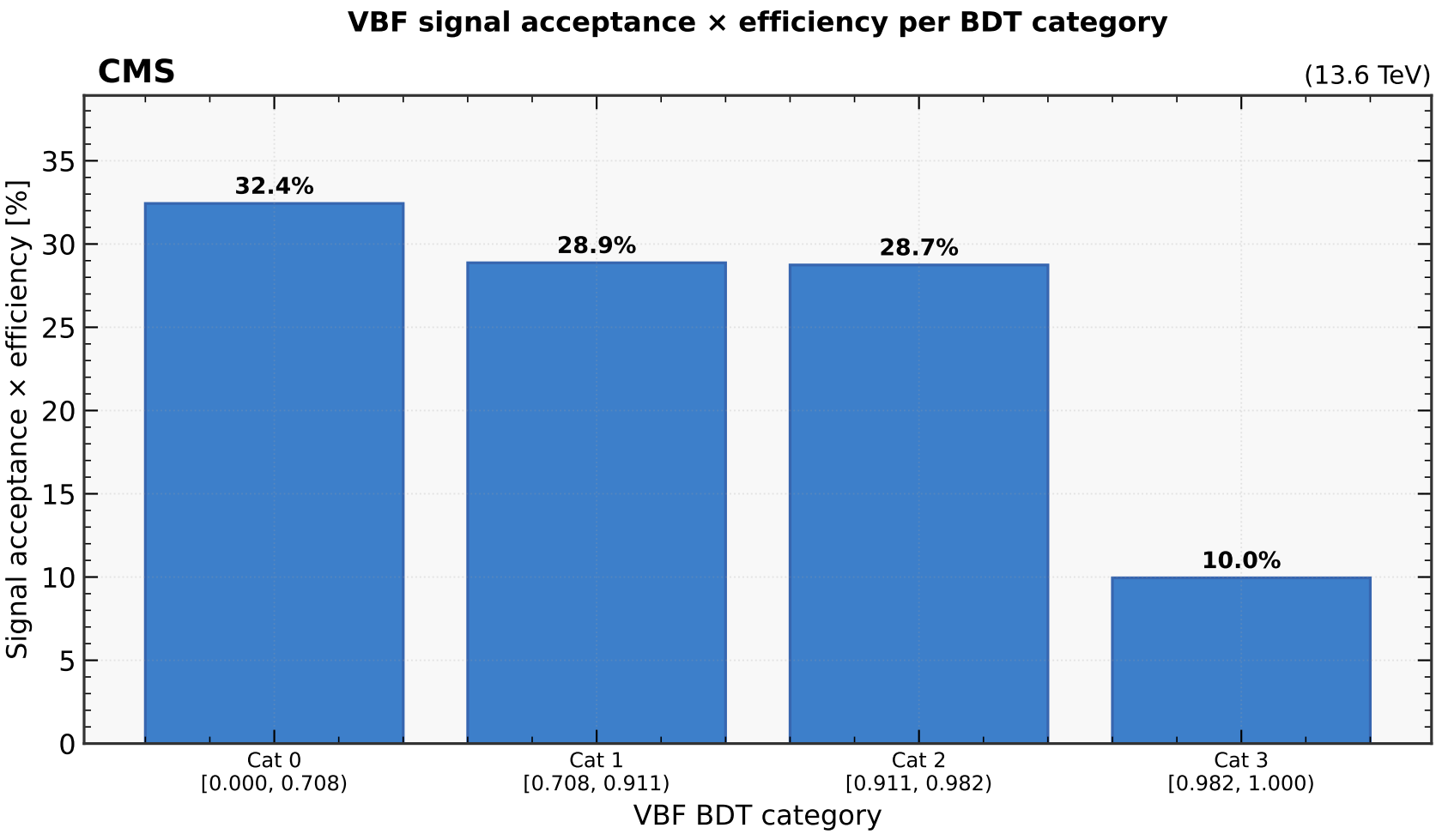




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.708)	20.42	124.7776	1.6271	1.5282	1.349	1.7686	4.883	3.832	0.145	0.001
1	[0.708, 0.911)	18.17	124.7855	1.6196	1.5418	1.450	1.8034	5.328	3.814	0.511	0.001
2	[0.911, 0.982)	18.09	124.7968	1.6023	1.5712	1.495	1.8218	5.955	3.773	0.994	0.001
3	[0.982, 1.000)	6.26	124.8176	1.5849	1.5970	1.617	1.8640	6.024	3.733	1.317	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.7
1.6
1.5
1.627
1.620
1.602
1.585
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
3.832
3.814
3.773
3.733
Cat 0 [0.000, 0.708)
Cat 1 [0.708, 0.911)
Cat 2 [0.911, 0.982)
Cat 3 [0.982, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
5
10
15
20
25
30
35
32.4%
28.9%
28.7%
10.0%
Cat 0
[0.000, 0.708)
Cat 1
[0.708, 0.911)
Cat 2
[0.911, 0.982)
Cat 3
[0.982, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.708)	20.42	92796.58	13.887	9109.102	[122.9, 126.7]	0.146	0.145
1	[0.708, 0.911)	18.17	6362.52	12.515	595.011	[122.9, 126.7]	0.513	0.511
2	[0.911, 0.982)	18.09	1601.29	12.553	155.380	[122.9, 126.7]	1.007	0.994
3	[0.982, 1.000)	6.26	127.81	4.398	9.785	[123.0, 126.7]	1.406	1.317

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.708)	14082	43161.17 ± 202.34	0.326	92796.58 ± 297.23	0.1455	0.1455	9109.102
1	[0.708, 0.911)	1149	2978.97 ± 50.60	0.386	6362.52 ± 73.96	0.5131	0.5113	595.011
2	[0.911, 0.982)	305	696.74 ± 23.03	0.438	1601.29 ± 35.78	1.0071	0.9939	155.380
3	[0.982, 1.000)	37	61.66 ± 6.30	0.600	127.81 ± 8.68	1.4061	1.3167	9.785
Combined	all						1.7332	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.708)	89373.93	329.97	2843.58	249.09	88347.52	92796.58
1	[0.708, 0.911)	5980.50	208.44	150.46	23.13	5469.67	6362.52
2	[0.911, 0.982)	1382.65	175.70	37.55	5.39	1280.43	1601.29
3	[0.982, 1.000)	81.12	43.85	2.58	0.26	75.33	127.81